

## **Assignment – I**

### **Mobile Application Development (TMC302)**

**Date 09-08-26, Last date 17-08-26**

**Q1.** Discuss the evolution of the Android Operating System. Explain the history, major milestones, and the key features that have made Android the most widely used mobile operating system.

**Q2.** Explain the Android Architecture with a neat labeled diagram. Describe the functions of each layer and explain how these layers interact to execute an Android application.

**Q3.** Android applications are composed of four major application components. Explain Activity, Service, Broadcast Receiver, and Content Provider with suitable real-life examples and practical use cases.

**Q4.** Describe the complete Activity Lifecycle with the help of a neat diagram. Explain the purpose of each lifecycle callback method and discuss its significance in Android application development.

**Q5.** Explain the Android Development Environment. Discuss the roles of JDK, JRE, JVM, Android SDK, and Android Studio, and illustrate how they work together during Android application development.

**Q6.** Differentiate between the following pairs with suitable examples:

- Android OS and Linux OS
- JDK and Android SDK
- Android Virtual Device (AVD) and Android Emulator
- Application Layer and Application Framework

**Q7.** Describe the complete procedure for creating the first Android application using Android Studio. Explain each project configuration option such as Project Name, Package Name, Language, and Minimum SDK, and justify why each is important.

**Q8.** Assume you have to develop an Android application but do not have access to a physical smartphone. Explain how Android Virtual Device (AVD) and the Android Emulator enable application development and testing. Discuss their advantages and limitations.

**Q9.** Why is Android Studio considered the official IDE for Android development? Explain its major features and discuss how these features improve application development, testing, debugging, and deployment.

**Q10.** A beginner wants to start Android application development from scratch. Explain, in sequence, the complete process from installing the development environment to successfully running the first Android application, highlighting the role of each software component involved.